

NATIONAL TEXTILE UNIVERSITY, FAISALABAD



House Price Prediction

DEPARTMENT OF COMPUTER SCIENCE

Course	Data Science
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Objective

The objective of this project is to develop a Machine Learning model that can accurately predict house selling prices based on various property features. The system helps estimate house prices by learning patterns from historical housing data and provides predictions through a user-friendly Streamlit web application.

2. Dataset Description

The project uses the Ames Housing Dataset, which contains information about residential properties such as house quality, living area, garage details, basement area, number of rooms, and other housing characteristics.

Target Variable

- SalePrice

Important Features

- OverallQual
- GrLivArea
- GarageCars
- GarageArea
- TotalBsmtSF
- 1stFlrSF
- FullBath
- TotRmsAbvGrd
- YearBuilt

3. Project Workflow

Step 1: Data Collection

- Load train.csv and test.csv datasets.

Step 2: Data Preprocessing

- Handle missing values.
- Convert categorical features into numerical format.
- Align train and test features.

Step 3: Feature Engineering

- Apply One-Hot Encoding on categorical features.
- Prepare data for machine learning.

Step 4: Train Test Splitting

- 80% data used for training.
- 20% data used for validation.

Step 5: Model Training

- Train Ridge Regression model.

Step 6: Model Evaluation

- Evaluate model using RMSE and R^2 Score.

Step 7: Model Saving

- Save trained model as model.pkl.
- Save feature names as features.pkl.

Step 8: Deployment

- Develop a Streamlit web application for real-time house price prediction.

4. Data Preprocessing Techniques

Handling Missing Values

Numerical Features

Missing numerical values were replaced using the median value of each column.

Reason

- Machine learning algorithms cannot process missing values.
- Median is less sensitive to outliers than mean.

Handling Categorical Missing Values

Missing categorical values were replaced with "None".

Reason

- Preserves data records.
- Treats missing values as a separate category.

One-Hot Encoding

Categorical features were converted into numerical format using One-Hot Encoding.

Example

Before Encoding:

Neighborhood

Downtown

Suburb

After Encoding:

Neighborhood_Downtown	Neighborhood_Suburb
1	0
0	1

Reason

Machine Learning models cannot understand text values directly.

Feature Alignment

Train and test datasets were aligned to ensure they contain identical columns.

Reason

Prevents feature mismatch during prediction.

Train-Test Split

The dataset was divided into:

- 80% Training Data
- 20% Validation Data

Reason

To evaluate model performance on unseen data.

5. Machine Learning Model

Model Used

Ridge Regression

Ridge Regression is an improved version of Linear Regression that adds regularization to reduce overfitting.

Why Ridge Regression?

- Fast and efficient.
- Suitable for regression problems.
- Handles large numbers of features.
- Reduces overfitting.
- Produces stable predictions.
- Performs well on housing price datasets.

Regularization

Ridge Regression applies a penalty to very large coefficients, helping the model generalize better on unseen data.

Model Performance

RMSE (Root Mean Squared Error)

RMSE: 28,660

Interpretation:

The model's predictions differ from actual house prices by approximately \$28,660 on average.

R² Score

R² Score: 0.8669

Interpretation:

The model explains approximately 86.69% of the variation in house prices.

Performance Summary

- R² Score = 86.69%
- RMSE = 28,660
- Model Performance = Very Good

7. Exploratory Data Analysis (EDA)

Four visualizations were created to understand the dataset and evaluate model performance.

Graph 1: Top Features with Missing Values

Purpose

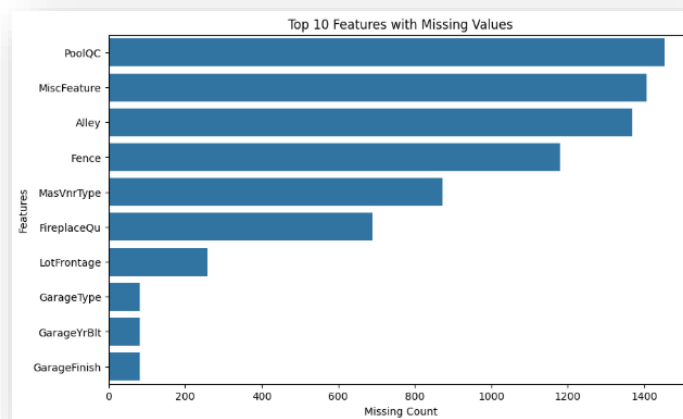
To identify features containing the highest number of missing values.

Findings

The following features contained significant missing data:

- PoolQC
- MiscFeature
- Alley
- Fence
- MasVnrType
- FireplaceQu
- LotFrontage

These features required preprocessing before model training.



Graph 2: Top Features Affecting House Price

Purpose

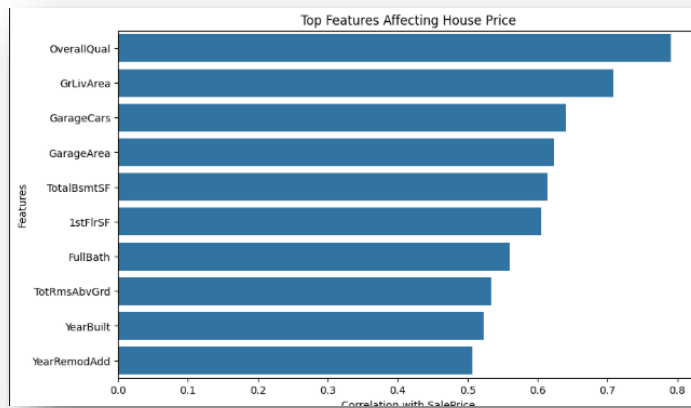
To identify the most important features influencing house prices.

Findings

The strongest features were:

- OverallQual
- GrLivArea
- GarageCars
- GarageArea
- TotalBsmtSF
- 1stFlrSF

These features showed strong positive relationships with SalePrice.



Graph 3: Correlation Heatmap

Purpose

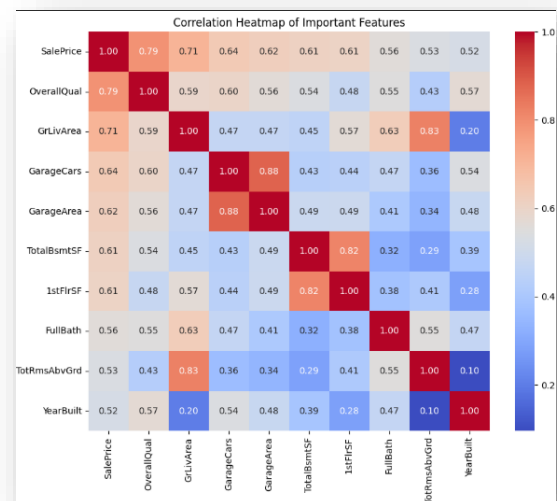
To analyze relationships among important features.

Findings

Strong correlations were observed between:

- GarageCars and GarageArea
- GrLivArea and TotRmsAbvGrd
- TotalBsmtSF and 1stFlrSF

This indicates that these features are closely related and significantly impact house prices.



Graph 4: Actual vs Predicted House Prices

Purpose

To evaluate model prediction accuracy.

Findings

Most data points were located near the diagonal reference line, indicating that predicted values are close to actual values.

This demonstrates that the model performs well and produces reliable predictions.

